TASK SPECIFICATION: P7-03 PHASE P7 — NOTIFICATIONS, GOVERNANCE & TELEMETRY ANALYTICS

Notification Center UI (Header Bell & Slide-out Drawer)

Git Branch: `feature/P7-03-notifications-ui` · Target Pull Request: feature/P7-03-notifications-ui

Core Objective & Technical Scope:

Top bar notification bell with animated unread badge counter and slide-out notification drawer with instant read actions.

Priority Level	Medium	Complexity Rating	S
Execution Status	Backlog 📋	Target PR / Commit	`feature/P7-03-notifications-ui`
Upstream Dependencies	P7-01		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Build NotificationBell component with animated unread counter.
2. Build NotificationDrawer displaying categorized notification cards.
3. Implement TanStack Query polling and optimistic mark-as-read mutations.
4. Add notification sound feedback.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `frontend/src/features/notifications/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Unread badge count synchronizes accurately with server state.
- Drawer slides smoothly with full keyboard accessibility.

VERIFICATION & TEST COVERAGE STRATEGY

- NotificationBell component tests
- NotificationDrawer interaction test